Archimedes Screw Lift

1. Number of Screw Revolutions Required

In an Archimedes screw mechanism, each full rotation lifts the object by a distance equal to the **pitch**.

Given:

Pitch = **1.2 mm**

Total lift length:

$$L = 255.203 + 70$$
$$L = 325.203\, mm$$

Number of revolutions:

$$N = \frac{L}{Pitch}$$
$$N = \frac{325.203}{1.2}$$
$$N \approx 271\, revolutions$$

Therefore, the screw must rotate **approximately 271 revolutions** to lift the ball from the bottom to the top.

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2. Relationship Between Motor Speed and Screw Speed

Since the motor shaft is directly connected to the screw shaft using a **Rigid Coupling**, the transmission ratio is **1:1**.

$$RPM_{motor} = RPM_{screw}$$

This means the screw rotates at the **same speed as the motor**.

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3. Time Required for Lifting

Case 1: Motor speed = 100 RPM

Convert RPM to revolutions per second:

$$100/60 = 1.67\,rps$$

Time required:

$$t = \frac{N}{rps}$$
$$t = \frac{271}{1.67}$$
$$t \approx 162\,s$$

Which is approximately:

2.7 minutes

Case 2: Motor speed = 200 RPM

Convert RPM to rps:

$$200/60 = 3.33\,rps$$

Time required:

$$t = \frac{271}{3.33}$$
$$t \approx 81\,s$$

Which is approximately:

1.3 minutes

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4. Torque Required to Rotate the Screw

Mass of the ball:

$$m = 0.015\,kg$$

Force required to lift the ball:

$$F = mg$$

$$F = 0.015 \times 9.81$$
$$F = 0.147\ N$$

Screw Dimensions

Screw diameter:

$$D = 27\ mm$$

Radius:

$$r = 13.5\ mm = 0.0135\ m$$

Torque Equation

$$T = \frac{F \times r}{\eta}$$

Where:

η = mechanical efficiency of the screw (≈0.6 for 3D printed parts)

$$T = \frac{0.147 \times 0.0135}{0.6}$$
$$T = 0.0033\ N \cdot m$$

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5. Effect of the Rigid Coupling

The **Rigid Coupling** transmits torque almost directly with very small losses.

Typical efficiency:

$$\eta \approx 0.98$$

Required motor torque:

$$T_{motor} = \frac{T}{0.98}$$
$$T_{motor} \approx 0.0034\ N \cdot m$$

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6. Safety Factor

To account for friction and design uncertainties, a safety factor of **3** is applied.

$$T_{required} = 0.0034 \times 3$$
$$T_{required} \approx 0.01\ N \cdot m$$

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7. Motor Selection

Recommended motor specifications:

Motor type: **Micro DC Gear Motor**

Voltage: **12 V**

Speed: **150 – 200 RPM**

Torque:

$$T \geq 0.1\ N \cdot m$$

This torque is significantly higher than the required torque, ensuring safe and reliable operation.

Electrical Components:

1. **12V N20 Micro Gear Motor (200 RPM)**
2. **Motor Driver Module (L298N)**
3. **Arduino Uno**
4. **Two IR sensors**
5. **12V DC Power Source**

Mechanical Components:

1. **Flexible Beam Coupling (3mm to 5mm)**

CODE

Lifting_Code | Arduino IDE 2.3.6

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Lifting_Code.ino

```
#define BottomSensor   2
#define TopSensor      3
#define motorPin1      8
#define motorPin2      7
#define enable         9
unsigned long timer = 0;
bool motorRunning = false;
void setup() {
  pinMode(BottomSensor, INPUT);
  pinMode(TopSensor, INPUT);
  pinMode(motorPin1, OUTPUT);
  pinMode(motorPin2, OUTPUT);
  pinMode(enable, OUTPUT);
}
void loop() {
  if(digitalRead(BottomSensor) == HIGH && !motorRunning){
    timer=millis();
    digitalWrite(motorPin1,HIGH);
    digitalWrite(motorPin2,LOW);
    analogWrite(enable,200);
    motorRunning = true;
  }
  if(digitalRead(TopSensor) == HIGH){
    digitalWrite(motorPin1,LOW);
    digitalWrite(motorPin2,LOW);
    analogWrite(enable,0);
    motorRunning = false;
  }
  if (millis() - timer > 170000 && motorRunning ) {
    digitalWrite(motorPin1, LOW);
    digitalWrite(motorPin2, LOW);
    analogWrite(enable, 0);
    motorRunning = false;
  }
}
```